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CSE A-2

Subject: Programming with Java

ASSIGNMENT No : 10

TITLE : Write a Java program to demonstrate the use of abstract classes and abstract methods

Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

Theory

Abstract classes provide partial implementation of a program and cannot be instantiated directly. They may contain both abstract and concrete methods. Abstract methods declare functionality that must be implemented by subclasses. Abstraction hides implementation details while exposing only essential functionality. This improves code flexibility and maintainability in object-oriented programming.

```
abstract class Shape {
    abstract void draw();

    void message () {
        System.out.println(x: "This is a shape");
    }
}

class Circle extends Shape {
    void draw() {
        System.out.println(x: "Drawing a circle.");
    }
}

public class Main {
    Run | Debug
    public static void main (String[] args) {
        Shape s = new Circle();
        s.draw();
        s.message();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp10> java Main.java
Drawing a circle.
This is a shape
```

Exercise

1. Create a Bank Account program where account number is final and cannot be changed.

```
abstract class Payment {
    abstract void pay(double amount);
    void display() {
        System.out.println(x: "Payment Method");
    }
}

class CreditCard extends Payment {
    void pay(double amount) {
        System.out.println("Paid Rs." + amount + " using Credit Card.");
    }
}

class UPI extends Payment {
    void pay(double amount) {
        System.out.println("Paid Rs." + amount + " using UPI.");
    }
}

public class Ex1 {
    Run | Debug
    public static void main(String[] args) {
        Payment p1 = new CreditCard();
        Payment p2 = new UPI();

        p1.display();
        p1.pay(amount: 2800);

        System.out.println();

        p2.display();
        p2.pay(amount: 1700);
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp10\Exercise> java Ex1.java
Payment Method
Paid Rs.2800.0 using Credit Card.

Payment Method
Paid Rs.1700.0 using UPI.
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp10\Exercise> █
```

2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
abstract class FoodOrder {
    double amount;
    FoodOrder(double amount) {
        this.amount = amount;
    }

    abstract void calculateBill();
}

class DineInOrder extends FoodOrder {
    DineInOrder(double amount) {
        super(amount);
    }

    void calculateBill() {
        double total = amount + 1000;
        System.out.println("Dine-In Bill : Rs." + total);
    }
}

class TakeAwayOrder extends FoodOrder {
    TakeAwayOrder(double amount) {
        super(amount);
    }

    void calculateBill() {
        double total = amount + 500;
        System.out.println("Take-Away Bill : Rs." + total);
    }
}

public class Ex2 {
    Run | Debug
    public static void main(String[] args) {
        FoodOrder order1 = new DineInOrder(amount: 1000);
        FoodOrder order2 = new TakeAwayOrder(amount: 1000);

        order1.calculateBill();
        order2.calculateBill();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp10\Exercise> java Ex2.java
Dine-In Bill : Rs.2000.0
Take-Away Bill : Rs.1500.0
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp10\Exercise> |
```